

Exercise and worked solution

Time series

Four parts to model time in finance: the moments of an AR(1), identifying a model from its signatures, diagnosing a unit root with the ADF, and measuring returns and their tails honestly. Work through each part before reading the solution.

Part A · Moments of an AR(1)

PROBLEM

A short rate is modelled by $Y_t = 0.9 + 0.7 Y_{t-1} + \varepsilon_t$, with shock variance $\sigma^2 = 1.8$. (a) Compute the long-run mean. (b) Compute the variance γ_0 . (c) Is the series stationary?

SOLUTION

1. (a) $\mu = \delta/(1-\phi) = 0.9/(1-0.7) = 0.9/0.3 = \mathbf{3}$. Classic trap: do not read the intercept 0.9 as the mean, persistence ϕ amplifies it.
2. (b) $\gamma_0 = \sigma^2/(1-\phi^2) = 1.8/(1-0.49) = 1.8/0.51 = \mathbf{3.53}$. The more persistent the process, the more volatile it is.
3. (c) Yes: $|\phi| = 0.7 < 1$, so the series is covariance-stationary and reverts to its long-run mean (mean reversion). At $\phi = 1$ it would become a random walk.

Part B · Identify the model

PROBLEM

You have three series. (i) ACF cutting off after lag 1, PACF decaying. (ii) ACF decaying, PACF cutting off after lag 2. (iii) ACF and PACF both decaying. Identify each model and say how you would estimate it.

SOLUTION

1. (i) ACF cutting off (lag 1) + PACF decaying = **MA(1)**. An MA component is present, the lagged shocks are not observable: estimate by **MLE**.
2. (ii) ACF decaying + PACF cutting off (lag 2) = **AR(2)**. Pure AR, observable regressors: estimate by **OLS** (biased in small samples but consistent).
3. (iii) Both decay = **ARMA**. MA component present: estimate by **MLE**. In all cases, then validate with a Ljung-Box on the residuals (they must be white noise).

Part C · Unit root and ADF test

PROBLEM

An ADF test with a constant ($n = 250$, 5% critical = -2.87) gives, on the level of a log index, a t of $\gamma = -1.80$. After moving to returns (log differences), the t is -9.1 . (a) Conclude at each step. (b) What is the order of integration? (c) Why not regress the levels on each other directly?

SOLUTION

1. (a) On the level: -1.80 is less negative than -2.87 , we **do not reject** $H_0 \rightarrow$ unit root, non-stationary series. On returns: $-9.1 < -2.87$ (far more negative), we **reject** $\rightarrow$ the returns are stationary.
2. (b) The series is **I(1)**: one differencing suffices to make it stationary (the norm for most asset prices).
3. (c) Regressing two unit-root level series produces a **spurious regression**: spectacular but meaningless R^2 and t , with a false-positive rate $> 75\%$. One must check stationarity and work on returns.

Part D · Returns, annualisation and tails

PROBLEM

(a) An asset's daily volatility is 1.2%. Annualise it (252 days, non-autocorrelated returns). (b) Why is the log return preferred for volatility models, while the simple return is used at portfolio level? (c) A Jarque-Bera test on daily returns gives a statistic of 480. What do you conclude (5% critical = 5.99)?

SOLUTION

1. (a) $\sigma_{\text{annual}} = \sqrt{252} \times 1.2\% \approx 15.87 \times 1.2\% = \mathbf{19.0\%}$. It is the variance that adds up ($\times 252$), so volatility grows in $\sqrt{252}$, not 252.
2. (b) The **log** return is additive *over time* (the multi-day return is the sum of daily returns), ideal for volatility models working at daily frequency. The **simple** return is additive *cross-sectionally*: a portfolio's return is the weighted average of its assets' simple returns.
3. (c) $480 \gg 5.99$, so we **reject normality** massively. Returns are fat-tailed (kurtosis > 3) and often skewed. Assuming normality would strongly underestimate tail risk (a loss $> 4\sigma$ is ~ 80 times more likely under a t_4 than under the normal).

Key takeaways: an AR(1) mean = $\delta/(1-\phi)$, stationary if $|\phi| < 1$; identify by ACF/PACF signatures (AR = PACF cuts off, MA = ACF cuts off, the mirror), estimate by OLS (pure AR) or MLE (once MA is present); the ADF tests $H_0 = \text{unit root}$ (left-tailed, Dickey-Fuller critical values); check stationarity before regressing to avoid the spurious; volatility in $\sqrt{h}$; and returns are fat-tailed (Jarque-Bera rejects normality).